

STANDARD OPERATING PROCEDURE

Banking Financial Data Processing Framework

Medallion Architecture Implementation

Document Classification: INTERNAL USE ONLY

Version: 1.0

Effective Date: January 15, 2026

Review Cycle: Quarterly

Owner: Data Engineering & Governance

Approval Authority: Chief Data Officer

1. PURPOSE AND SCOPE

1.1 Purpose

This Standard Operating Procedure (SOP) establishes **mandatory** protocols for all data processing activities within the Banking Financial Services data lake environment. Compliance is **non-negotiable**. Any deviation from these procedures requires written approval from the Chief Data Officer and must be documented in the exception log.

1.2 Scope

This SOP applies to:

- All data engineering personnel with data lake access
- ETL/ELT pipeline developers and maintainers
- Data quality assurance teams
- Business intelligence and analytics consumers
- Third-party vendors processing BFS data

1.3 Regulatory Context

This framework ensures compliance with:

- Basel III/IV capital adequacy requirements
- SOX Section 404 (Internal Controls)
- GDPR Article 32 (Security of Processing)
- PCI-DSS Requirements 3.4 and 10.2
- Local banking regulations and data residency laws

Violation of these procedures may result in regulatory penalties, disciplinary action, or termination.

2. MEDALLION ARCHITECTURE OVERVIEW

2.1 Architecture Mandate

The data lake **SHALL** be organized using the three-tier Medallion Architecture. This structure is **not optional** and deviations will not be tolerated.

2.2 Layer Definitions

2.2.1 Bronze Layer (Raw Data Zone)

Purpose: Immutable landing zone for raw source data

Mandatory Characteristics:

- Data ingested in native format with **zero transformations**
- Full historical retention (minimum 7 years per regulatory requirements)
- Append-only operations—deletions are **strictly prohibited**
- Complete lineage metadata recorded at ingestion
- Encryption at rest using AES-256 standard

Naming Convention:

`bronze/{source_system}/{schema}/{table}/year={YYYY}/month={MM}/day={DD}/`

Storage Format: Parquet or Delta Lake (mandatory for transactional data)

2.2.2 Silver Layer (Validated Data Zone)

Purpose: Cleansed, conformed, and validated enterprise data

Mandatory Characteristics:

- All data quality rules applied and enforced
- Business keys validated against master data
- Type conversions, standardizations, and deduplication completed
- Audit columns added (processing_timestamp, dq_status, source_record_hash)
- Failed records quarantined with complete rejection metadata

Naming Convention:

silver/{domain}/{entity}/year={YYYY}/month={MM}/day={DD}/

Quality Threshold: Minimum 99.5% pass rate required for production promotion

2.2.3 Gold Layer (Business-Ready Zone)

Purpose: Aggregated, enriched data optimized for consumption

Mandatory Characteristics:

- Dimensional models and aggregated fact tables only
- Conformed dimensions across business units
- Pre-calculated metrics and KPIs
- Optimized for query performance (indexed, partitioned, cached)
- Documentation embedded in metadata catalog

Naming Convention:

gold/{business_domain}/{data_product}/version={v#}/

Access Control: Role-based access with documented business justification

3. DATA INGESTION PROTOCOLS

3.1 Source System Requirements

All source systems **MUST**:

- Provide full extract plus incremental change data capture (CDC)
- Include data lineage identifiers in every record
- Timestamp all records with source system time (ISO 8601 format)
- Implement retry logic with exponential backoff
- Maintain idempotency for all data transfers

3.2 Ingestion Frequency Standards

Data Classification	Maximum Latency	Ingestion Method
Critical Financial Transactions	15 minutes	Real-time streaming (Kafka/Kinesis)
Regulatory Reporting Data	1 hour	Micro-batch CDC

Data Classification	Maximum Latency	Ingestion Method
Customer Master Data	4 hours	Incremental batch
Reference Data	Daily	Full extract at 02:00 UTC
Historical Archives	Monthly	Scheduled bulk transfer

These timelines are binding service level agreements (SLAs). Breaches must be escalated immediately.

3.3 Mandatory Validation Steps

Before data enters Bronze layer:

1. Schema validation against registered data contracts
2. File integrity checks (checksum/hash verification)
3. Record count reconciliation with source system
4. Duplicate detection within batch
5. Malware scanning for file-based ingestion

Validation failures result in automatic quarantine. Do not override.

4. DATA TRANSFORMATION REQUIREMENTS

4.1 Bronze to Silver Transformation Rules

You MUST implement the following in sequence:

1. Data Quality Enforcement

- Null handling according to approved business rules
- Data type validation and casting
- Range and boundary checks
- Referential integrity validation
- Format standardization (dates, currencies, addresses)

2. Business Logic Application

- Currency conversions using official exchange rates (loaded daily)
- Timezone normalization to UTC
- PII masking for non-production environments
- Calculated fields per approved calculation registry

3. Audit Trail Creation

- Record-level hash for change detection
- Processing timestamp (nanosecond precision)
- Data quality score (0-100 scale)
- Transformation version identifier

4. Error Handling Protocol

- Critical errors: Reject record, log to quarantine table
- Warning errors: Accept with flag, alert data steward
- Information: Accept, record in processing log

Transformation code MUST be version-controlled and peer-reviewed before production deployment.

4.2 Silver to Gold Transformation Rules

Aggregation Standards:

- Use approved aggregation functions only (documented in data dictionary)
- Maintain grain consistency within subject areas
- Apply slowly changing dimension (SCD) logic Type 2 for historical tracking
- Pre-aggregate to approved reporting granularities

Performance Mandates:

- Partition by date for time-series data
- Z-order/cluster by primary query dimensions
- Maintain table statistics current (refresh weekly minimum)
- Implement compaction on schedule (Delta Lake OPTIMIZE)

5. DATA QUALITY FRAMEWORK

5.1 Quality Dimensions (Non-Negotiable)

All data **WILL** be assessed against six quality dimensions:

1. **Completeness:** Required fields populated (target: 100%)
2. **Accuracy:** Values match source of truth (target: 99.9%)
3. **Consistency:** Cross-field validation rules satisfied (target: 99.5%)
4. **Timeliness:** Data available within SLA window (target: 99.9%)
5. **Validity:** Values within acceptable domain ranges (target: 100%)

6. **Uniqueness:** No duplicate records on business keys (target: 100%)

5.2 Quality Gates

Bronze to Silver Promotion:

- Automated quality checks pass at 99.5% threshold
- Schema drift alerts resolved
- Reconciliation with source system complete

Silver to Gold Promotion:

- Business rule validation 100% compliant
- Cross-table referential integrity verified
- Performance benchmarks met (query response < 3 seconds)

If quality gates fail, pipelines WILL halt. Manual override requires VP-level approval and incident documentation.

5.3 Data Quality Monitoring

Real-time Monitoring Requirements:

- Continuous profiling of incoming data streams
- Anomaly detection using statistical process control
- Automated alerting for quality threshold breaches
- Dashboard visibility for data stewards

Reporting Cadence:

- Daily quality scorecards to data owners
- Weekly trend analysis to governance committee
- Monthly quality KPI reporting to executive leadership

6. SECURITY AND ACCESS CONTROL

6.1 Encryption Standards

At Rest:

- AES-256 encryption mandatory for all layers
- Key rotation every 90 days (automated)

- Hardware security module (HSM) for key management

In Transit:

- TLS 1.3 minimum for all data movement
- Certificate-based authentication required
- No plaintext transmission permitted

6.2 Access Control Matrix

Layer	Default Access	Approval Required	Audit Frequency
Bronze	Data Engineering Only	Manager	Monthly
Silver	Engineering + Analytics	Director	Monthly
Gold	Business Users (RBAC)	Data Owner	Quarterly

Principle of Least Privilege is absolute. Access requests require documented business justification.

6.3 Data Classification Handling

Highly Confidential Data (PII, account numbers):

- Tokenization or encryption required in Silver/Gold
- Access logged with user identity and query
- Production data prohibited in non-production environments
- Masking functions applied for development/testing

Confidential Data (aggregated financials):

- Row-level security enforced
- Column-level masking where appropriate
- Access granted by data owner approval

Internal Use Data:

- Standard RBAC controls
 - Aggregated only, no individual identifiers
-

7. PIPELINE ORCHESTRATION STANDARDS

7.1 Workflow Requirements

All pipelines MUST:

- Implement idempotent operations (rerunnable without side effects)
- Include comprehensive logging (INFO level minimum)
- Provide alerting for failures (PagerDuty integration)
- Support backfill operations without data corruption
- Include rollback procedures in runbook

7.2 Scheduling Mandates

Production Pipelines:

- Scheduled via enterprise orchestrator (Airflow/Databricks Workflows)
- Dependency management explicit (upstream/downstream)
- Service windows respected (no processing during month-end close)
- Retry logic: Maximum 3 attempts with 5-minute intervals

Critical Path Pipelines:

- Real-time monitoring with 99.9% uptime SLA
- Automated failover to secondary processing cluster
- Incident response within 15 minutes

7.3 Change Management

All pipeline changes require:

1. Change request ticket (minimum 5 business days notice)
2. Impact analysis documentation
3. Peer code review with approval
4. Testing in non-production environment
5. Rollback plan documented
6. Change advisory board approval for production

Emergency changes permitted only for severity-1 incidents with retrospective documentation.

8. METADATA AND LINEAGE MANAGEMENT

8.1 Metadata Registration Requirements

Every data asset MUST have:

- Business name and technical name
- Data owner and data steward assignment
- Business glossary term mappings
- Data classification level
- Retention policy and archival rules
- Quality rules and validation logic
- Sample data and usage examples

Metadata registry is the single source of truth. Unregistered datasets are unauthorized.

8.2 Lineage Tracking Mandates

End-to-end lineage SHALL capture:

- Source system and extraction timestamp
- All transformation logic applied
- Data quality checkpoints and outcomes
- Column-level lineage for calculated fields
- Consumption patterns and downstream dependencies

Lineage must be queryable and visualizable within the data catalog.

8.3 Documentation Standards

Required Documentation:

- Data flow diagrams (updated within 2 weeks of changes)
- Transformation business logic specifications
- Data dictionary with field definitions
- Processing schedules and dependencies
- Incident runbooks and escalation procedures

Outdated documentation is a compliance violation.

9. MONITORING AND OBSERVABILITY

9.1 Monitoring Requirements

Pipeline Health Metrics:

- Execution duration (baseline and threshold)
- Data volume processed (with anomaly detection)
- Error rates and failure patterns
- Resource utilization (CPU, memory, I/O)

Data Quality Metrics:

- Quality dimension scores by layer
- Quarantine volumes and rejection reasons
- Data freshness and latency measurements
- Schema evolution tracking

9.2 Alerting Configuration

Critical Alerts (Immediate Response):

- Pipeline failures on critical path
- Data quality below 95% threshold
- Security policy violations
- Data loss or corruption detected

Warning Alerts (Response within 4 hours):

- Processing delays exceeding SLA by 20%
- Quality scores between 95-99%
- Unusual data volume patterns

Informational Alerts (Next business day review):

- Successful backfill completions
- Schema changes detected
- New data source registrations

9.3 Incident Management

Severity Definitions:

Severity 1: Production data unavailable, regulatory reporting at risk

Response Time: 15 minutes

Resolution Target: 4 hours

Severity 2: Non-critical pipeline failure, degraded performance

Response Time: 1 hour

Resolution Target: 24 hours

Severity 3: Minor issues, no business impact

Response Time: 4 hours

Resolution Target: 5 business days

All incidents require root cause analysis and corrective action plan within 48 hours of resolution.

10. DISASTER RECOVERY AND BUSINESS CONTINUITY

10.1 Backup Requirements

Bronze Layer:

- Full backup weekly, incrementals daily
- Retention: 7 years minimum
- Geographic replication to secondary region

Silver/Gold Layers:

- Incremental backups daily
- Retention: 3 years minimum
- Cross-region replication for gold layer

Recovery Time Objectives (RTO):

- Critical pipelines: 4 hours
- Standard pipelines: 24 hours
- Archive data: 72 hours

Recovery Point Objectives (RPO):

- Real-time data: 15 minutes
- Batch data: 1 hour
- Reference data: 24 hours

10.2 Failover Procedures

Automated Failover Conditions:

- Primary region unavailable for 15 minutes
- Data corruption detected in primary
- Security breach in primary environment

Manual Failover Procedures:

- Documented in disaster recovery runbook
 - Tested quarterly (mandatory)
 - Executive approval required for production failover
-

11. COMPLIANCE AND AUDIT

11.1 Audit Trail Requirements

System logs **MUST** capture:

- All data access (user, timestamp, data accessed)
- All data modifications (before/after values)
- All pipeline executions (parameters, outcomes)
- All security events (authentication, authorization)

Log Retention: 7 years, immutable storage

11.2 Compliance Reporting

Monthly Reports:

- Data quality scorecard
- SLA compliance metrics
- Security and access reviews
- Change management summary

Quarterly Reports:

- Comprehensive audit of data lineage
- Policy compliance assessment
- Risk register updates
- Capacity planning analysis

11.3 Audit Cooperation

During regulatory audits:

- Provide complete data lineage documentation within 24 hours
- Grant auditor access to read-only production replicas
- Demonstrate control effectiveness with evidence
- Document all audit findings and remediation plans

Non-compliance with audit requests is grounds for immediate escalation to legal counsel.

12. ROLES AND RESPONSIBILITIES

12.1 Data Engineering Team

- Build and maintain pipelines per this SOP
- Ensure data quality standards met
- Respond to pipeline incidents per SLA
- Document all processes and changes

12.2 Data Governance Office

- Enforce policy compliance
- Approve exceptions to procedures
- Conduct quality audits
- Maintain data catalog and glossary

12.3 Data Owners

- Define business rules and validation logic
- Approve access requests
- Review quality reports
- Participate in incident reviews

12.4 Information Security

- Enforce encryption and access controls
- Conduct security assessments
- Investigate security incidents

- Approve privileged access
-

13. TRAINING AND COMPETENCY

13.1 Mandatory Training

All personnel MUST complete:

- Data handling and classification training (annually)
- Security awareness certification (annually)
- This SOP acknowledgment (quarterly)
- Technology-specific certifications (per role)

Training completion is tracked and enforced. Non-compliance blocks system access.

13.2 Competency Assessment

Before production access granted:

- Pass technical assessment (85% minimum score)
 - Complete supervised work period (30 days)
 - Demonstrate SOP compliance in practice
 - Manager certification of competency
-

14. CHANGE CONTROL FOR THIS SOP

14.1 Review and Update Process

This SOP SHALL be reviewed:

- Quarterly by Data Governance Committee
- After any regulatory change
- Following significant incidents
- Upon technology platform changes

14.2 Version Control

All changes require:

- Change request with business justification
- Impact analysis

- Stakeholder review period (15 business days)
- Chief Data Officer approval
- Communication to all affected personnel

Version history maintained in document management system.

15. ACKNOWLEDGMENT AND ENFORCEMENT

15.1 Acknowledgment Requirement

All covered personnel MUST:

- Read this SOP in full
- Complete comprehension assessment
- Sign acknowledgment form
- Renew acknowledgment quarterly

Access to data lake systems is contingent upon current acknowledgment.

15.2 Enforcement and Consequences

Violations are classified as:

Minor: Documentation errors, missed reporting deadlines

Consequence: Written warning, remedial training

Major: Quality gate overrides, unauthorized access, incomplete testing

Consequence: Formal reprimand, access suspension, performance impact

Critical: Data breaches, regulatory violations, intentional non-compliance

Consequence: Immediate access revocation, disciplinary action up to termination, potential legal action

There are no exceptions. Compliance is mandatory.

APPENDICES

Appendix A: Glossary of Terms

Appendix B: Data Quality Rule Catalog

Appendix C: Pipeline Runbook Templates

Appendix D: Incident Response Flowcharts

Appendix E: Access Request Form

Appendix F: Exception Request Form

Appendix G: Technology Stack Reference

Appendix H: Regulatory Requirement Mapping

DOCUMENT CONTROL

Prepared By: Data Engineering & Governance Team

Reviewed By: Chief Data Officer, Chief Information Security Officer, General Counsel

Approved By: Chief Data Officer

Effective Date: January 15, 2026

Next Review Date: April 15, 2026

This is a controlled document. Printed copies are uncontrolled and may be outdated.

ACKNOWLEDGMENT STATEMENT

I acknowledge that I have read, understood, and agree to comply with all provisions of this Standard Operating Procedure. I understand that violations may result in disciplinary action up to and including termination of employment or contract.

Name: Dheemanth Dev

Signature: _____

Date: 15 January 2026

Manager Approval: Aswini Roy